

Assignment 4: Classification of Textual Data

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Abstract

In this assignment, we investigated the comparative performance of a Long Short-Term Memory (LSTM) network versus a fine-tuned BERT model on the classification of scientific abstracts using the Web of Science (WOS-11967) dataset. We implemented the LSTM from scratch using pre-trained GloVe embeddings to establish a sequential baseline and compared it against the pre-trained `bert-base-uncased` transformer. We found that the fine-tuned BERT approach achieved consistently better accuracy than the LSTM across both classification levels. Specifically, BERT achieved a test accuracy of 93.6% on the seven level 1 scientific domains, outperforming the LSTM's 88.6%. On the more granular task of classifying 34 specific sub-fields, the performance gap widened, with BERT achieving 87.6% accuracy compared to 75.0% for the LSTM. These findings suggest that the self-attention mechanisms and pre-training of BERT provide a significant advantage in capturing the nuanced semantic context required for distinguishing closely related scientific disciplines.

1 Introduction

For this assignment, we evaluated the performance of a Long-Short Term Memory model and a fine-tuned BERT model on the Web of Science WOS-11967 dataset. The WOS-11967 dataset consists of 11,967 abstracts labeled with the scientific field and sub-field they belong to [1]. We quantified the ability of our models to successfully predict the scientific field and sub-field that each abstract belongs to, addressing both the 7-class parent task (Level 1) and the more granular 34-class child task (Level 2). To contrast traditional sequential modeling with modern transfer learning, we implemented the LSTM architecture from scratch using pre-trained GloVe embedding, which allows the model to leverage global co-occurrence statistics to capture semantic relationships between scientific terms, avoiding the sparsity issues of one-hot encoding [2]. Our experiments demonstrate that the pre-trained BERT model outperforms the LSTM baseline across both tasks. This performance gap underscores the effectiveness of transfer learning and self-attention mechanisms in capturing the intricate semantic context of scientific literature.

2 Datasets

In this assignment, we used the Web of Science WOS-11967 dataset. This dataset comprises 11,967 abstracts, each labeled with both the scientific field and the specific subfield to which it belongs. There are seven total scientific field labels: Computer Science, Electrical Engineering, Psychology, Mechanical Engineering, Civil Engineering, Medical Science, and Biochemistry. Within these parent categories, a total of 34 sub-specialties were identified across all fields. To ensure rigorous evaluation, the data was stratified into Training (72%), Validation (8%), and Test (20%) sets using the Level 2 labels to preserve the distribution of minority classes. To dive deeper into the dataset, we performed exploratory analysis to help us choose our hyperparameter values.

An exploratory analysis of the class distributions revealed an incoherence in the level 2 labels. In fact, we noticed that the YL2.txt labels contained only relative indexes that depended on the YL1.txt labels and that the global Y.txt labels contained a class mismatch for label 10. We decided to create global indices for the level 2 labels; specifically, we analyzed how many unique parent-child label combinations existed within the dataset. By concatenating the Level 1 field ID with the Level 2 sub-field ID, we identified 34 distinct semantic categories. We then mapped these combinations to a new global integer range [0, 33], resolving the ambiguity in the raw data and establishing a valid 34-class classification target for our experiments.

After building our labels encoding, we performed an analysis of the level 1 labels, which revealed a slight imbalance in classes with the majority class appearing approximately twice as often as the minority class (Figure 1a). For level 2, the imbalance was definitely more skewed; specifically, we identified Class 7 as an extreme outlier with fewer than 100 samples, suggesting the models would likely struggle to generalize on this specific sub-field (Figure 1b).

To optimize our dataset structures, we analyzed token count per abstract and found an average length of 195 words. Based on this, we established a maximum sequence length of 256, which covers 83.71% of the dataset without truncation while adhering to memory constraints (Figure 1c).

Ultimately, we performed word frequency analysis in order to filter unnecessary words. The top 20 words revealed a dominance of generic academic terms (e.g., "study," "results") and functional stopwords (Figure 1d). Motivated by this, we initially attempted a preprocessing pipeline using NLTK to remove stopwords and increase the semantic density of the input vectors. However, experimental runs showed that this aggressive filtering removed necessary syntactic context, leading to optimization instability and gradient divergence in the LSTM. Consequently, we reverted to a standard Regex-based tokenizer for the final baseline to ensure training stability.

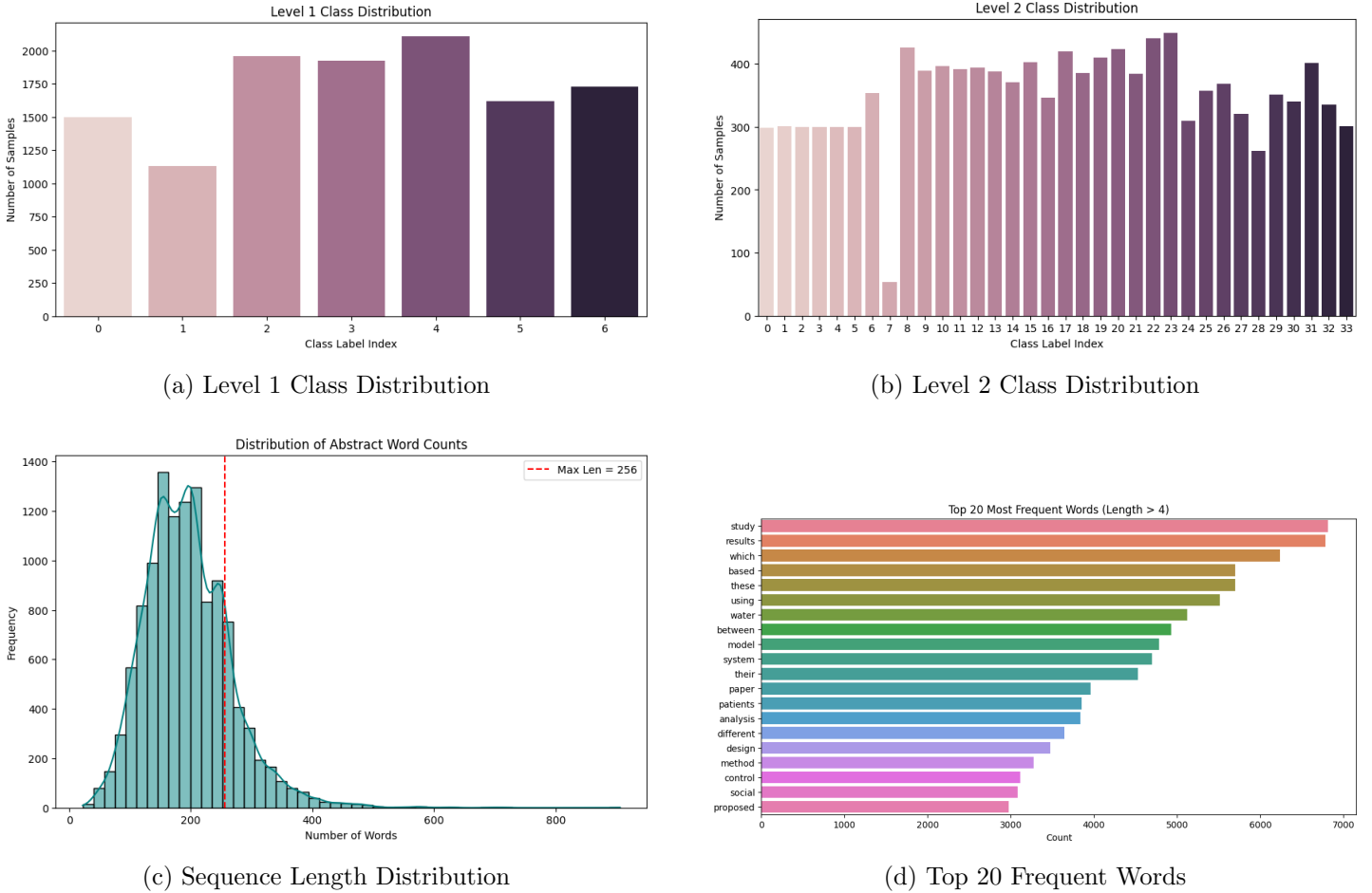


Figure 1: Exploratory Data Analysis of WOS-11967.

3 Results

3.1 Model Performance Comparison

Table 1 summarizes the classification accuracy for both models across the Level 1 (7 classes) and Level 2 (33 classes) tasks. The fine-tuned BERT model consistently outperformed the LSTM baseline, achieving a peak accuracy of **93.6%** on the Level 1 task as compared to the LSTM with an accuracy of 88.6%, an improvement of 5%. Furthermore, we see further improvement on the secondary classification task with the fine-tuned BERT model achieving an accuracy of 87.55%, a 12.98% improvement over the LSTM which has an accuracy of 74.98%.

Task	Model	Accuracy
Level 1 (7 Classes)	LSTM	0.8860
Level 1 (7 Classes)	BERT	0.9361
Level 2 (33 Classes)	LSTM	0.7498
Level 2 (33 Classes)	BERT	0.8755

Table 1: Test set accuracy comparing LSTM (trained from scratch) and BERT (fine-tuned) on hierarchical classification tasks.

We found that the pre-trained BERT model consistently outperformed our LSTM model on both classification tasks. There are several factors that may explain this improvement. A BERT model is based on the transformer architecture, which leverages self-attention to better understand text by analyzing pairwise context between every pair of words. This allows for better comprehension of context as compared to LSTM models which process sequentially, and thus may dilute the influence of context between distant words.

Furthermore, our pre-trained BERT model also carries the advantages of transfer learning. As the model learns on a larger generalized dataset, it is able to better resist overfitting on smaller datasets. Furthermore, it is able to learn fundamental language features such as sentence structure, grammar, and common words before adapting to WOS-11967 through fine-tuning. Furthermore, fine-tuning is comparatively less expensive than the complete training of an LSTM, reducing training time and lowering computational resources required. These advantages are why our pre-trained BERT model consistently outperforms our LSTM model on both our classification tasks.

3.2 Qualitative Analysis and Creativity

3.2.1 Interpretability: Attention Mechanisms

To validate that the fine-tuned BERT model leverages semantic context, we visualized the self-attention weights from the last transformer block (Layer 12, Head 1). Figure 2 contrasts the model’s focus during a correct versus an incorrect prediction.

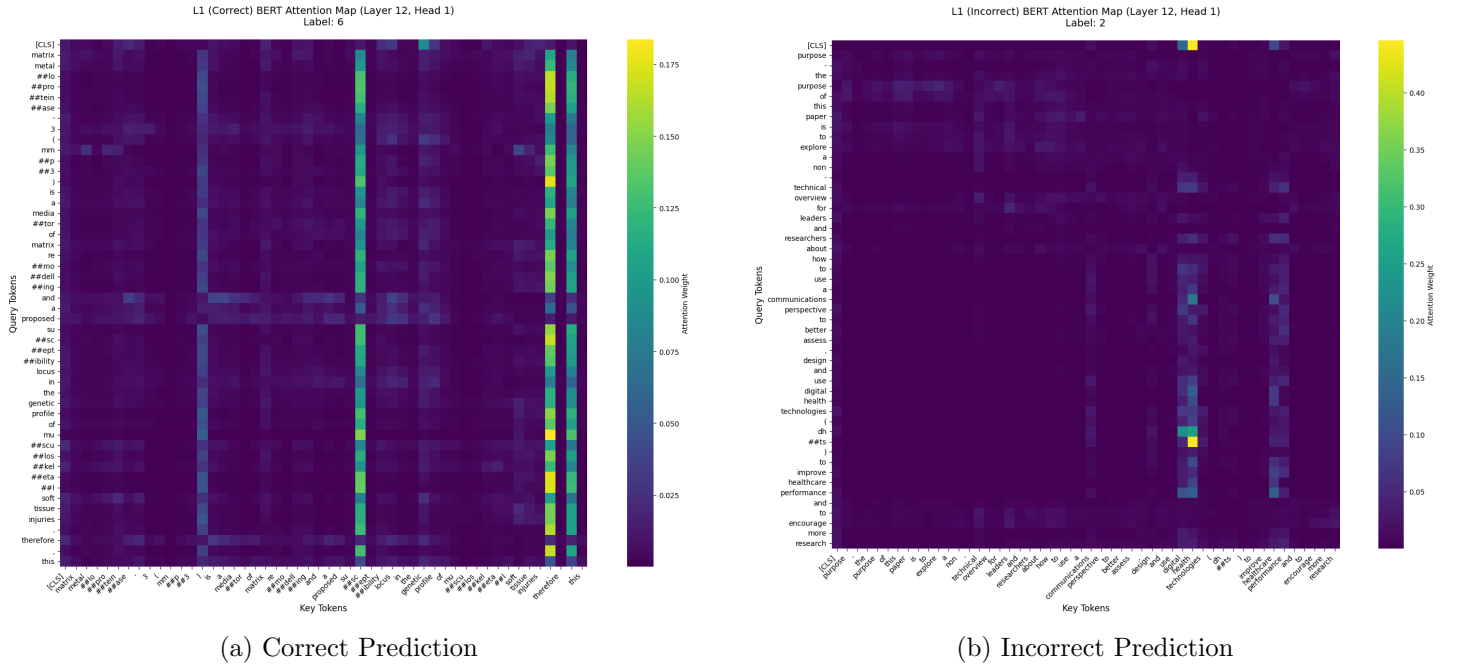


Figure 2: BERT Self-Attention Maps.

From Figure 2, we see how when making a correct prediction, our BERT model has high attention between our query classification token [CLS] and domain-specific key tokens such as "genetic," and "locus," but does not show very high attention with any specific key token. Furthermore, we also see that connector words such as "therefore" and "this" have high attention with most words. In contrast, our incorrect prediction has very high attention

between query [CLS] and key "health," indicating that it was possibly misled by this outlier and overestimated its importance, resulting in an incorrect prediction.

3.2.2 Creativity: Error Analysis

As a creative extension to the required metrics, we computed a Confusion Matrix to diagnose specific failure modes (Figure 3). This analysis reveals that misclassifications are not random; they cluster among semantically related fields. In fact, the most misclassified pair was Medical Science and Biochemistry, as their semantic vocabularies are closer together than any other field.

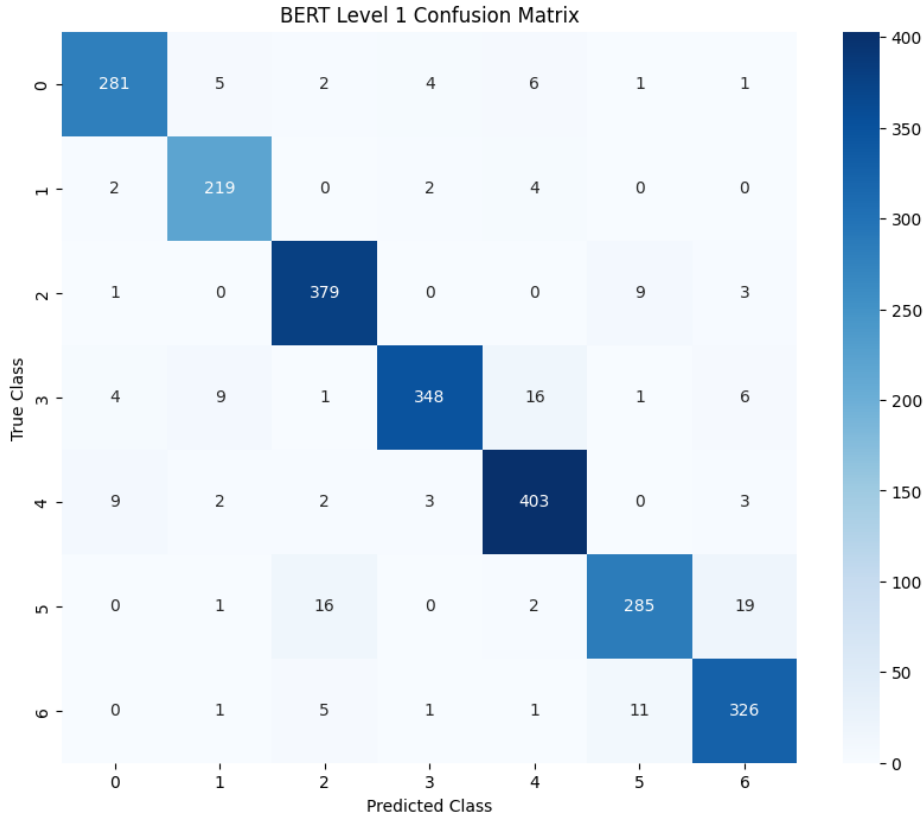


Figure 3: Confusion Matrix for BERT (Level 1).

3.2.3 Training Time Analysis

We additionally explore the training time of our pre-trained BERT and LSTM models, as seen in Figure 4. We note that the pre-trained BERT model takes significantly less epochs to approach asymptotic loss, taking only around 5 epochs compared to the LSTM’s 25 epochs on both tasks. This is because the model is pre-trained, so the only training that needs to be done is fine-tuning of the model for our specific classification task, highlighting another advantage of transfer learning.

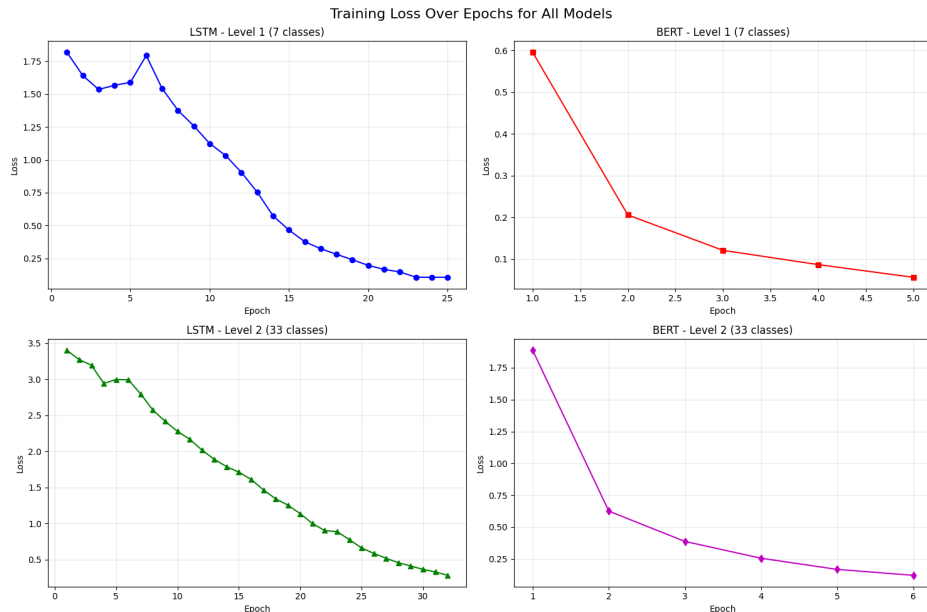


Figure 4: Training loss over epochs for BERT and LSTM models with early stopping.

4 Discussion and Conclusion

In this project, we compare the performance of an LSTM model and a fine-tuned BERT model on the classification task of identifying primary field of research and subfield from an abstract using the Web of Science WOS-11967 dataset. After initial dataset exploratory analysis

We find that the fine-tuned BERT model consistently outperforms the LSTM model. For the primary field classification task, BERT achieves a test accuracy of 93.61%, surpassing the LSTM’s 88.60%. On the more complex subfield classification task, the BERT shows a greater improvement, achieving an accuracy 87.55% compared to the LSTM’s 74.98%. This discrepancy is likely due to the LSTM’s inability to retain long-term information, whereas the transformer’s self-attention mechanism allows understanding of global context regardless of distance.

Beyond performance, we also analyze the interpretability, errors, and training efficiency of our BERT model. We find that the BERT model successfully pays attention domain-specific keywords (e.g., "genetic", "locus") while filtering out noise, while incorrect predictions may result from excessive attention paid to outlier tokens. Through error analysis via confusion matrices, we find that errors occur most often between related fields, such as between Medical Science and Biochemistry. Furthermore, the pre-trained BERT model converges significantly faster, reaching asymptotic loss in about 5 epochs versus 25 for the LSTM. This confirms the value of transfer learning, as the model is able to use prior knowledge of syntax and structure and require only fine-tuning to the specific task as compared to LSTM, which must learn fundamental language information from scratch.

Future investigation could explore performance differences between BERT and LSTM models on larger inputs such as entire papers, where the advantage of global context may be more apparent and the BERT model may be able to better distinguish between related fields. Furthermore, exploration of a BERT model trained from scratch may help in better highlighting the advantage of transfer learning.

5 Statement of Contributions

Vladimir developed the BERT and LSTM models. Samy added an exploratory analysis of our dataset, hyperparameter tuning, early stopping, and our confusion matrix, and helped write the report. Alan helped write the report.

References

- [1] K. Kowsari, D. E. Brown, M. Heidarysafa, K. Jafari Meimandi, , M. S. Gerber, and L. E. Barnes, “Hdltex: Hierarchical deep learning for text classification,” in *Machine Learning and Applications (ICMLA), 2017 16th IEEE International Conference on*. IEEE, 2017.
- [2] J. Pennington, R. Socher, and C. D. Manning, “Glove: Global vectors for word representation,” in *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, 2014, pp. 1532–1543. [Online]. Available: <https://aclanthology.org/D14-1162>